

UNIT-5: SAMPLING AND SAMPLING DISTRIBUTION

A Comprehensive Theoretical, Mathematical, and Practical Guide with Extended Problem Sets

Course Content Outline

1. Conceptual Foundations of Sampling and Populations
2. Types of Sampling: Probability and Non-Probability Methods
3. Sampling Error and the Central Limit Theorem (CLT)
4. Point and Interval Estimation: Theoretical Framework
5. Confidence Limits for Population Mean (Large & Small Samples)
6. Confidence Limits for Population Proportion
7. Confidence Limits for the Difference of Two Independent Means
8. Confidence Limits for the Difference of Two Proportions
9. Comprehensive Problem Sets & Practical Solutions

1. Conceptual Foundations of Sampling and Populations

1.1 Basic Terminology

In quantitative analysis, statistical inference allows researchers to draw conclusions about vast groups based on observations from a subset. To understand these processes, several fundamental terms must be rigorously defined:

- **Population (Universe):** The entire collection of items, individuals, or measurements under consideration. It represents the complete set of observations that possess a common specified characteristic. A population can be finite (e.g., all registered voters in a city) or infinite (e.g., all possible outcomes of rolling a pair of dice indefinitely).
- **Sample:** A representative subset of the population selected for investigation. The sample size is denoted by the symbol n .
- **Parameter:** A descriptive numerical measure computed from an entire population. Parameters are typically constant but unknown values, denoted by Greek letters. Common examples include the population mean (μ), population standard deviation (σ), and population proportion (P).
- **Statistic:** A descriptive numerical measure computed from sample data. Statistics are random variables whose values depend on the specific sample drawn, denoted by Roman letters. Examples include the sample mean ($\bar{x}$), sample standard deviation (s), and sample proportion ($\hat{p}$).

1.2 Parameters vs. Statistics Summary

Measure	Population Parameter (Greek)	Sample Statistic (Roman)
Mean	μ	$\bar{x}$
Variance	σ^2	s^2 (or S^2 modified)
Standard Deviation	σ	s
Proportion	P or π	p or $\hat{p}$
Size	N	n

1.3 The Rationale for Sampling

Conducting a complete census of a population is often impractical or completely impossible due to several constraints:

1. **Economic Constraints:** Gathering data from every single member of a population requires massive budgets. Sampling delivers highly accurate results at a small fraction of the cost.
2. **Temporal Constraints:** Time is a critical asset. Census collection can take months or years, rendering data obsolete by publication. Sampling offers rapid insights.
3. **Destructive Testing:** In manufacturing quality control (e.g., testing the breaking strength of cables or the lifespan of batteries), testing a unit completely destroys it. A complete census would leave nothing left to sell.
4. **Physical Impossibility:** Infinite populations (e.g., deep-sea biological metrics or continuous factory output) cannot be fully measured.

2. Types of Sampling: Probability and Non-Probability Methods

Sampling methodologies are categorized into two primary frameworks: Probability Sampling and Non-Probability Sampling.

2.1 Probability Sampling Methods

Probability sampling requires that every element in the population has a known, non-zero chance of being selected. This allows for rigorous statistical analysis and margin-of-error calculations.

2.1.1 Simple Random Sampling (SRS)

In SRS, each member of the population has an equal probability of selection, and every possible combination of n elements has an equal chance of being chosen. It can be performed in two ways:

- **Simple Random Sampling With Replacement (SRSWR):** An item is drawn, recorded, and returned to the population before the next draw. The population size remains N for every draw. The total number of unique samples possible is N^n .
- **Simple Random Sampling Without Replacement (SRSWOR):** An item is drawn and excluded from all future draws. The population size shrinks by one at each stage. The total number of unique samples possible is given by the combination formula: ${}^N C_n = N! / (n!(N-n)!)$.

2.1.2 Stratified Random Sampling

When a population is heterogeneous but contains distinct internal subgroups (strata) that are internally homogeneous, stratified sampling increases precision. The population N is divided into non-overlapping strata $N_1, N_2, \dots, N_k$, and independent simple random samples are drawn from each stratum. Allocation can be **proportional** (sample sizes relative to stratum sizes: $n_i = n * (N_i/N)$) or **optimum** (Neyman allocation, which balances both size and internal variance).

2.1.3 Systematic Sampling

In systematic sampling, elements are selected at a regular, fixed interval from an ordered list. The sampling interval is calculated as:

$$k = N / n$$

A random starting number between 1 and k is chosen. Let this be r . The subsequent items selected are $r, r+k, r+2k, r+3k$, and so on.

2.1.4 Cluster Sampling

When the population is naturally divided into geographic or structural units called clusters (e.g., schools, city blocks), cluster sampling can be highly efficient. Instead of sampling individual elements, researchers select a random sample of whole clusters. All individuals within the chosen clusters are then examined.

2.2 Non-Probability Sampling Methods

Non-probability sampling selects elements based on subjective judgment or convenience rather than random chance. These methods do not allow for formal probability-based margins of error.

- **Convenience Sampling:** Selecting units that are easiest to reach or access (e.g., interviewing people outside a local mall).
- **Judgmental (Purposive) Sampling:** The researcher uses their personal expertise to handpick individuals who they believe best fit the focus of the study.
- **Quota Sampling:** The population is categorized into specific groups, and the researcher collects data from a set number of individuals in each group using convenience methods.
- **Snowball Sampling:** Used for hard-to-reach populations. Current participants recruit additional subjects from their own networks.

3. Sampling Error and the Central Limit Theorem (CLT)

3.1 Understanding Sampling Distributions

Because a sample statistic (such as $\bar{x}$) depends entirely on which specific elements are randomly drawn, it is a random variable. The probability distribution of a sample statistic across all possible samples of size n is called its **Sampling Distribution**.

The standard deviation of a sample statistic's distribution is specifically called its **Standard Error (SE)**. It measures how much the statistic varies across different samples.

3.2 The Central Limit Theorem (CLT)

The Central Limit Theorem is the foundation of inferential statistics. It states that:

The Central Limit Theorem: Given a population with any arbitrary probability distribution shape, having a finite mean μ and a finite standard deviation σ , the sampling distribution of the sample mean $\bar{x}$ approaches a perfect normal distribution as the sample size n becomes large (typically $n \geq 30$).

Mathematically, the sampling distribution of the mean has:

- **Expected Value (Mean):** $E(\bar{x}) = \mu_{\bar{x}} = \mu$
- **Standard Error (Infinite or With Replacement):** $SE(\bar{x}) = \sigma_{\bar{x}} = \sigma / \sqrt{n}$
- **Standard Error (Finite Without Replacement):** If sampling without replacement from a finite population of size N , we apply the Finite Population Correction (FPC) factor:

$$\sigma_{\bar{x}} = (\sigma / \sqrt{n}) * \sqrt{(N - n) / (N - 1)}$$

3.3 Standard Error Formulas for Various Statistics

Statistic	Standard Error Formula (Infinite / SRSWR)	Standard Error Formula (Finite SRSWOR)
Sample Mean ($\bar{x}$)	$\sigma / \sqrt{n}$	$(\sigma / \sqrt{n}) * \sqrt{((N-n)/(N-1))}$
Sample Proportion (p)	$\sqrt{(P(1-P) / n)}$	$\sqrt{(P(1-P) / n)} * \sqrt{((N-n)/(N-1))}$
Difference of Two Means ($\bar{x}_1 - \bar{x}_2$)	$\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$	-
Difference of Two Proportions ($p_1 - p_2$)	$\sqrt{(P_1Q_1/n_1 + P_2Q_2/n_2)}$	-

4. Point and Interval Estimation: Theoretical Framework

Statistical estimation uses sample data to approximate unknown population parameters. It is divided into two primary approaches: Point Estimation and Interval Estimation.

4.1 Point Estimation

A point estimation calculates a single numerical value from the sample to serve as the best guess for the unknown parameter. For instance, using the sample mean $\bar{x} = 45.2$ to estimate the population mean μ .

Criteria for a Good Estimator:

1. **Unbiasedness:** An estimator is unbiased if its expected value equals the true population parameter.

$$E(\bar{x}) = \mu \Rightarrow \bar{x} \text{ is an unbiased estimator of } \mu$$

Note that the sample variance using divisor n is biased. The unbiased sample variance s^2 requires a divisor of $n - 1$:

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

2. **Consistency:** As the sample size n increases toward infinity, the value of the estimator converges closer to the true population parameter. The standard error approaches zero.
3. **Efficiency:** If two estimators are both unbiased, the one with the smaller variance across samples is considered more efficient.
4. **Sufficiency:** An estimator is sufficient if it utilizes all the information about the parameter contained within the sample data.

4.2 Interval Estimation

A point estimate provides no information about its precision or margin of error. An interval estimation addresses this by defining a range of values within which the true population parameter is expected to fall with a specified level of confidence.

The general structural formula for an interval estimate is:

$$\text{Point Estimate} \pm (\text{Reliability Factor} \times \text{Standard Error})$$

The **Confidence Level**, denoted as $(1 - \alpha)$, represents the long-term probability that the interval construction procedure will contain the true parameter. α is the significance level (the probability that the interval misses the parameter).

Common Critical Values from the Standard Normal Z-Distribution:

- For a 90% Confidence Level ($\alpha = 0.10$): $Z_{\alpha/2} = 1.645$
- For a 95% Confidence Level ($\alpha = 0.05$): $Z_{\alpha/2} = 1.96$
- For a 99% Confidence Level ($\alpha = 0.01$): $Z_{\alpha/2} = 2.576$

5. Confidence Limits for Population Mean

Constructing an interval estimate for a population mean μ depends on sample size (n) and whether the population standard deviation (σ) is known.

5.1 Case 1: Large Sample Size ($n \geq 30$) or Known σ

When $n \geq 30$, the Central Limit Theorem justifies using the standard normal Z-distribution, even if the underlying population distribution is unknown. If the population standard deviation σ is unknown, we substitute the sample standard deviation s .

The formula for the confidence limits is:

$$\bar{x} \pm Z_{\alpha/2} * (\sigma / \sqrt{n})$$

5.2 Case 2: Small Sample Size ($n < 30$) and Unknown σ

If the sample size is small ($n < 30$), the population is normally distributed, and σ is unknown, we use the **Student's t-distribution** instead of the Z-distribution. The t-distribution is wider, accounting for the extra uncertainty of estimating σ with s .

The degrees of freedom (df) for a single sample mean are:

$$df = n - 1$$

The formula for the confidence limits is:

$$\bar{x} \pm t_{\alpha/2, df} * (s / \sqrt{n})$$

Where the unbiased sample standard deviation s is calculated as:

$$s = \sqrt{ \sum (x - \bar{x})^2 / (n - 1) }$$

6. Confidence Limits for Population Proportion

When dealing with categorical attributes (success/failure, yes/no), we estimate the population proportion P using the sample proportion p .

The sample proportion is calculated as:

$$p = x / n$$

Where x is the number of successes observed in the sample of size n . Let $q = 1 - p$ represent the sample failure proportion.

6.1 Large Sample Approximation for Proportions

According to the normal approximation to the binomial distribution, if both $n * p \geq 5$ and $n * q \geq 5$ (some guidelines require ≥ 10), the sampling distribution of p is approximately normal. Its standard error is estimated using sample metrics:

$$SE(p) = \sqrt{(p * q / n)}$$

The formula for the confidence limits for the population proportion P is:

$$p \pm Z_{\alpha/2} * \sqrt{(p(1 - p) / n)}$$

6.2 Finite Population Adjustment for Proportions

If the sample is drawn from a finite population of size N without replacement, and the sample size is a significant portion of the population ($n/N > 0.05$), the standard error must include the correction factor:

$$SE(p) = \sqrt{(p * q / n)} * \sqrt{((N - n) / (N - 1))}$$

7. Confidence Limits for the Difference of Two Independent Means

To compare two distinct populations (such as comparing salaries between two industries), we estimate the difference between their population means: $\mu_1 - \mu_2$.

7.1 Case 1: Large Independent Samples ($n_1 \geq 30, n_2 \geq 30$)

With large sample sizes, the sampling distribution of the difference between the two sample means ($\bar{x}_1 - \bar{x}_2$) is approximately normal. If the population variances σ_1^2 and σ_2^2 are unknown, we substitute the sample variances s_1^2 and s_2^2 .

The standard error formula is:

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{(\sigma_1^2 / n_1 + \sigma_2^2 / n_2)}$$

The confidence limits are calculated as:

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} * \sqrt{(s_1^2 / n_1 + s_2^2 / n_2)}$$

7.2 Case 2: Small Independent Samples with Equal Variances

If $n_1 < 30$ or $n_2 < 30$, the populations are normally distributed, and we assume the unknown variances are equal ($\sigma_1^2 = \sigma_2^2$), we pool the sample variances to calculate a more accurate shared variance estimate (s_p^2):

$$s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / [n_1 + n_2 - 2]$$

The standard error using this pooled estimate is:

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{(s_p^2 / n_1 + s_p^2 / n_2)} = s_p * \sqrt{(1/n_1 + 1/n_2)}$$

The degrees of freedom for the t-distribution are:

$$df = n_1 + n_2 - 2$$

The confidence limits are:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} * s_p * \sqrt{1/n_1 + 1/n_2}$$

8. Confidence Limits for the Difference of Two Proportions

To compare the proportion of a specific attribute across two independent populations (e.g., success rates of two different treatments), we estimate the difference between their population proportions: $P_1 - P_2$.

8.1 Large Independent Samples Formula

When sample sizes are sufficiently large, the sampling distribution of the difference between two sample proportions ($p_1 - p_2$) is approximately normal. The standard error is calculated as:

$$SE(p_1 - p_2) = \sqrt{[(p_1 * q_1 / n_1) + (p_2 * q_2 / n_2)]}$$

Where:

- p_1 is the sample proportion from population 1, and $q_1 = 1 - p_1$
- p_2 is the sample proportion from population 2, and $q_2 = 1 - p_2$

The confidence limits for $P_1 - P_2$ are calculated as:

$$(p_1 - p_2) \pm Z_{\alpha/2} * \sqrt{[(p_1(1 - p_1) / n_1) + (p_2(1 - p_2) / n_2)]}$$

8.2 Summary Matrix for Confidence Interval Formulas

Target Parameter	Sample Size / Condition	Distribution	Standard Error (SE)
Mean (μ)	Large ($n \geq 30$) or σ known	Normal (Z)	$\sigma / \sqrt{n}$ (or $s / \sqrt{n}$)
Mean (μ)	Small ($n < 30$) & σ unknown	Student's t ($df=n-1$)	$s / \sqrt{n}$
Proportion (P)	Large ($np \geq 5, nq \geq 5$)	Normal (Z)	$\sqrt{pq / n}$
Difference in Means ($\mu_1 - \mu_2$)	Large ($n_1, n_2 \geq 30$)	Normal (Z)	$\sqrt{s_1^2/n_1 + s_2^2/n_2}$
Difference in Means ($\mu_1 - \mu_2$)	Small ($n_1, n_2 < 30$), Equal Var	Student's t ($df=n_1+n_2-2$)	$s_p \times \sqrt{(1/n_1 + 1/n_2)}$
Difference in Proportions ($P_1 - P_2$)	Large samples	Normal (Z)	$\sqrt{p_1q_1/n_1 + p_2q_2/n_2}$

9. Comprehensive Problem Sets & Practical Solutions

This section provides detailed step-by-step solutions for a wide variety of practical applications across engineering, finance, quality control, and social sciences.

PROBLEM 1: SAMPLING DISTRIBUTION OF THE MEAN (INFINITE POPULATION)

An industrial bottling plant fills juice bottles. The volume of juice filled per bottle follows a distribution with a known mean of 500 mL and a standard deviation of 12 mL. A quality control inspector randomly samples 36 bottles from the production line. What is the probability that the sample mean volume falls between 496 mL and 503 mL?

Solution:

Step 1: Identify given parameters.

Population mean (μ) = 500 mL

Population standard deviation (σ) = 12 mL

Sample size (n) = 36

Step 2: Calculate the Standard Error of the Mean.

Since the sample size $n = 36 \geq 30$, according to the Central Limit Theorem, the sampling distribution of the sample mean $\bar{x}$ is approximately normal.

$$SE(\bar{x}) = \sigma_{\bar{x}} = \sigma / \sqrt{n} = 12 / \sqrt{36} = 12 / 6 = 2.0 \text{ mL}$$

Step 3: Convert raw boundaries into Standard Normal Z-scores.

$$\text{Formula: } Z = (\bar{x} - \mu) / \sigma_{\bar{x}}$$

For $\bar{x} = 496 \text{ mL}$:

$$Z_1 = (496 - 500) / 2.0 = -4 / 2.0 = -2.00$$

For $\bar{x} = 503 \text{ mL}$:

$$Z_2 = (503 - 500) / 2.0 = 3 / 2.0 = 1.50$$

Step 4: Find probabilities from standard normal tables.

We want to find $P(-2.00 \leq Z \leq 1.50)$.

$$P(-2.00 \leq Z \leq 1.50) = P(Z \leq 1.50) - P(Z \leq -2.00)$$

From Z-tables:

$$P(Z \leq 1.50) = 0.9332$$

$$P(Z \leq -2.00) = 0.0228$$

Subtracting the values:

$$\text{Probability} = 0.9332 - 0.0228 = 0.9104$$

Conclusion: There is a 91.04% probability that the sample mean volume of the 36 bottles will fall between 496 mL and 503 mL.

PROBLEM 2: SAMPLING DISTRIBUTION OF THE MEAN WITH FINITE POPULATION CORRECTION

A corporate department has exactly 200 servers. The average uptime for these servers is 120 hours per week with a standard deviation of 15 hours. A technician randomly selects a sample of 50 servers without replacement. Find the mean and standard error of the sampling distribution of the sample mean uptime.

Solution:

Step 1: Identify given parameters.

Total Population Size (N) = 200

Sample Size (n) = 50

Population Mean (μ) = 120 hours

Population Standard Deviation (σ) = 15 hours

Step 2: Check the sampling fraction.

$$n / N = 50 / 200 = 0.25$$

Since $0.25 > 0.05$ (25% of the population is sampled), we must apply the Finite Population Correction (FPC) factor to calculate the standard error.

Step 3: Calculate the mean of the sampling distribution.

$$\mu_{\bar{x}} = \mu = 120 \text{ hours}$$

Step 4: Calculate the Standard Error with FPC.

$$\text{Formula: } \sigma_{\bar{x}} = (\sigma / \sqrt{n}) * \sqrt{(N - n) / (N - 1)}$$

First, compute the standard error without correction:

$$\sigma / \sqrt{n} = 15 / \sqrt{50} = 15 / 7.0711 = 2.1213$$

Next, compute the FPC factor:

$$FPC = \sqrt{(200 - 50) / (200 - 1)} = \sqrt{150 / 199} = \sqrt{0.7538} = 0.8682$$

Multiply the two values:

$$\sigma_{\bar{x}} = 2.1213 * 0.8682 = 1.8416 \text{ hours}$$

Conclusion: The mean of the sampling distribution is 120 hours, and the standard error is 1.84 hours.

PROBLEM 3: CONFIDENCE INTERVAL FOR MEAN (LARGE SAMPLE, KNOWN VARIANCE)

A shipping logistics firm wants to estimate the average weight of packages delivered daily. A random sample of 100 packages yields a sample mean weight of 5.4 kg. The historical population standard deviation is known to be 1.2 kg. Construct a 95% confidence interval for the true population mean weight of all packages.

Solution:

Step 1: Identify given parameters.

Sample size (n) = 100 (Large sample, $n \geq 30$)

Sample mean ($\bar{x}$) = 5.4 kg

Population standard deviation (σ) = 1.2 kg

Confidence Level = 95% ($\alpha = 0.05$)

Step 2: Determine the critical value.

For a two-tailed 95% confidence interval, the critical value from the standard normal Z-table is:

$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

Step 3: Calculate the Standard Error of the Mean.

$$SE(\bar{x}) = \sigma / \sqrt{n} = 1.2 / \sqrt{100} = 1.2 / 10 = 0.12 \text{ kg}$$

Step 4: Calculate the Margin of Error.

$$\text{Margin of Error (ME)} = Z_{\alpha/2} * SE(\bar{x}) = 1.96 * 0.12 = 0.2352 \text{ kg}$$

Step 5: Compute the Confidence Limits.

$$\text{Lower Limit} = \bar{x} - ME = 5.4 - 0.2352 = 5.1648 \text{ kg}$$

$$\text{Upper Limit} = \bar{x} + ME = 5.4 + 0.2352 = 5.6352 \text{ kg}$$

Conclusion: We are 95% confident that the true population mean weight of all packages lies between 5.16 kg and 5.64 kg.

PROBLEM 4: CONFIDENCE INTERVAL FOR MEAN (LARGE SAMPLE, UNKNOWN VARIANCE)

An accountant examines the processing time for expense invoices. A random sample of 64 invoices reveals a sample mean processing time of 24.5 minutes with a sample standard deviation of 4.8 minutes. Construct a 99% confidence interval for the true mean processing time of all invoices.

Solution:

Step 1: Identify given parameters.

Sample size (n) = 64 (Large sample, $n \geq 30$)

Sample mean ($\bar{x}$) = 24.5 minutes

Sample standard deviation (s) = 4.8 minutes

Confidence Level = 99% ($\alpha = 0.01$)

Step 2: Determine the critical value.

For a two-tailed 99% confidence interval, the critical Z-value is:

$$Z_{\alpha/2} = Z_{0.005} = 2.576$$

Step 3: Calculate the Standard Error.

Since the population variance σ is unknown, we substitute the sample standard deviation s :

$$SE(\bar{x}) \approx s / \sqrt{n} = 4.8 / \sqrt{64} = 4.8 / 8 = 0.60 \text{ minutes}$$

Step 4: Calculate the Margin of Error.

$$ME = Z_{\alpha/2} * SE(\bar{x}) = 2.576 * 0.60 = 1.5456 \text{ minutes}$$

Step 5: Compute the Confidence Limits.

$$\text{Lower Limit} = 24.5 - 1.5456 = 22.9544 \text{ minutes}$$

$$\text{Upper Limit} = 24.5 + 1.5456 = 26.0456 \text{ minutes}$$

Conclusion: We are 99% confident that the true population mean invoice processing time is between 22.95 minutes and 26.05 minutes.

PROBLEM 5: CONFIDENCE INTERVAL FOR MEAN (SMALL SAMPLE, T-DISTRIBUTION)

A concrete manufacturer measures the compressive strength of a new composite mixture. A random sample of 10 structural blocks is tested, yielding the following strengths in megapascals (MPa): 42, 45, 41, 44, 46, 43, 40, 45, 42, 44. Construct a 95% confidence interval for the mean compressive strength of the population, assuming a normal distribution.

Solution:

Step 1: Compute sample metrics (mean and standard deviation).

Sum of values (Σx) = 42 + 45 + 41 + 44 + 46 + 43 + 40 + 45 + 42 + 44 = 432

Sample Mean ($\bar{x}$) = 432 / 10 = 43.2 MPa

Now, calculate the sum of squared deviations from the mean to find the sample variance:

$\Sigma(x_i - \bar{x})^2$:

$$(42 - 43.2)^2 = (-1.2)^2 = 1.44$$

$$(45 - 43.2)^2 = (1.8)^2 = 3.24$$

$$(41 - 43.2)^2 = (-2.2)^2 = 4.84$$

$$(44 - 43.2)^2 = (0.8)^2 = 0.64$$

$$(46 - 43.2)^2 = (2.8)^2 = 7.84$$

$$(43 - 43.2)^2 = (-0.2)^2 = 0.04$$

$$(40 - 43.2)^2 = (-3.2)^2 = 10.24$$

$$(45 - 43.2)^2 = (1.8)^2 = 3.24$$

$$(42 - 43.2)^2 = (-1.2)^2 = 1.44$$

$$(44 - 43.2)^2 = (0.8)^2 = 0.64$$

Sum of deviations = 1.44 + 3.24 + 4.84 + 0.64 + 7.84 + 0.04 + 10.24 + 3.24 + 1.44 + 0.64 = 34.4

Unbiased Sample Variance (s^2) = 34.4 / (10 - 1) = 34.4 / 9 = 3.8222

Sample Standard Deviation (s) = $\sqrt{3.8222}$ = 1.9550 MPa

Step 2: Determine degrees of freedom and critical t-value.

$$df = n - 1 = 10 - 1 = 9$$

For a 95% confidence level and 9 degrees of freedom, looking at a two-tailed t-table:

$$t_{0.025, 9} = 2.262$$

Step 3: Calculate the Standard Error of the Mean.

$$SE(\bar{x}) = s / \sqrt{n} = 1.9550 / \sqrt{10} = 1.9550 / 3.1623 = 0.6182 \text{ MPa}$$

Step 4: Calculate the Margin of Error.

$$ME = t_{\alpha/2, df} * SE(\bar{x}) = 2.262 * 0.6182 = 1.3984 \text{ MPa}$$

Step 5: Compute the Confidence Limits.

$$\text{Lower Limit} = 43.2 - 1.3984 = 41.8016 \text{ MPa}$$

$$\text{Upper Limit} = 43.2 + 1.3984 = 44.5984 \text{ MPa}$$

Conclusion: We are 95% confident that the true population mean compressive strength of the concrete mixture is between 41.80 MPa and 44.60 MPa.

PROBLEM 6: CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

A digital marketplace platform surveys a random sample of 400 active consumers and finds that 120 of them use mobile digital wallets for checkouts. Construct a 95% confidence interval for the true population proportion of all users who prefer mobile digital wallets.

Solution:

Step 1: Compute sample proportions.

Sample size (n) = 400

Number of successes (x) = 120

Sample proportion (p) = $120 / 400 = 0.30$

Sample failure proportion (q) = $1 - 0.30 = 0.70$

Step 2: Verify normality assumptions.

$$n * p = 400 * 0.30 = 120 \geq 10$$

$$n * q = 400 * 0.70 = 280 \geq 10$$

The sample satisfies the conditions for the normal approximation.

Step 3: Determine the critical Z-value.

For a 95% confidence level, $Z_{\alpha/2} = 1.96$.

Step 4: Calculate the Standard Error of the Proportion.

$$SE(p) = \sqrt{(p * q / n)} = \sqrt{(0.30 * 0.70 / 400)} = \sqrt{(0.21 / 400)} = \sqrt{0.000525} = 0.02291$$

Step 5: Calculate the Margin of Error.

$$ME = Z_{\alpha/2} * SE(p) = 1.96 * 0.02291 = 0.0449$$

Step 6: Compute the Confidence Limits.

$$\text{Lower Limit} = 0.30 - 0.0449 = 0.2551$$

$$\text{Upper Limit} = 0.30 + 0.0449 = 0.3449$$

Conclusion: We are 95% confident that the true population proportion of consumers who use mobile digital wallets for checkouts is between 25.51% and 34.49%.

PROBLEM 7: CONFIDENCE INTERVAL FOR THE DIFFERENCE BETWEEN TWO INDEPENDENT MEANS (LARGE SAMPLES)

A manufacturing group evaluates the operational lifespans of two machinery component brands. Brand A: $n_1 = 50$, sample mean ($\bar{x}_1$) = 1250 hours, sample standard deviation (s_1) = 80 hours.

Brand B: $n_2 = 60$, sample mean ($\bar{x}_2$) = 1190 hours, sample standard deviation (s_2) = 95 hours.

Construct a 99% confidence interval for the true difference between the mean operational lifespans of the two brands ($\mu_1 - \mu_2$).

Solution:

Step 1: Identify given parameters.

Brand A: $n_1 = 50$, $\bar{x}_1 = 1250$, $s_1 = 80$

Brand B: $n_2 = 60$, $\bar{x}_2 = 1190$, $s_2 = 95$

Confidence Level = 99% ($\alpha = 0.01$)

Step 2: Determine the critical Z-value.

For a 99% confidence level, $Z_{\alpha/2} = 2.576$.

Step 3: Compute the point estimate for the difference.

$\bar{x}_1 - \bar{x}_2 = 1250 - 1190 = 60$ hours

Step 4: Calculate the Standard Error of the Difference of Means.

$$SE = \sqrt{s_1^2 / n_1 + s_2^2 / n_2}$$

$$SE = \sqrt{80^2 / 50 + 95^2 / 60}$$

$$SE = \sqrt{6400 / 50 + 9025 / 60}$$

$$SE = \sqrt{128 + 150.4167} = \sqrt{278.4167} = 16.6858 \text{ hours}$$

Step 5: Calculate the Margin of Error.

$$ME = Z_{\alpha/2} * SE = 2.576 * 16.6858 = 42.9826 \text{ hours}$$

Step 6: Compute the Confidence Limits.

$$\text{Lower Limit} = 60 - 42.9826 = 17.0174 \text{ hours}$$

$$\text{Upper Limit} = 60 + 42.9826 = 102.9826 \text{ hours}$$

Conclusion: We are 99% confident that the true difference between the mean lifespans of Brand A and Brand B is between 17.02 hours and 102.98 hours. Because this interval does not contain zero, we can conclude that Brand A components have a significantly longer average lifespan.

PROBLEM 8: CONFIDENCE INTERVAL FOR DIFFERENCE OF MEANS (SMALL SAMPLES, POOLED VARIANCE)

Two groups of commercial real estate appraisers use different evaluation models to value properties. The variances of both approaches are assumed to be equal. Group 1: $n_1 = 8$, sample mean ($\bar{x}_1$) = \$320k, sample standard deviation (s_1) = \$18k.

Group 2: $n_2 = 10$, sample mean ($\bar{x}_2$) = \$305k, sample standard deviation (s_2) = \$15k.

Construct a 95% confidence interval for the difference between the mean valuations of the two groups ($\mu_1 - \mu_2$).

Solution:

Step 1: Compute the pooled variance estimate.

$$s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / [n_1 + n_2 - 2]$$

$$s_p^2 = [(8 - 1)(18)^2 + (10 - 1)(15)^2] / [8 + 10 - 2]$$

$$s_p^2 = [7 * 324 + 9 * 225] / 16 = [2268 + 2025] / 16 = 4293 / 16 = 268.3125$$

$$s_p = \sqrt{268.3125} = 16.3802 \text{ thousand dollars}$$

Step 2: Determine degrees of freedom and critical t-value.

$$df = n_1 + n_2 - 2 = 8 + 10 - 2 = 16$$

Looking at a two-tailed t-table for 16 degrees of freedom at a 95% confidence level ($\alpha = 0.05$):

$$t_{0.025, 16} = 2.120$$

Step 3: Calculate the Standard Error of the Difference.

$$SE = s_p * \sqrt{(1/n_1 + 1/n_2)} = 16.3802 * \sqrt{(1/8 + 1/10)} = 16.3802 * \sqrt{(0.125 + 0.10)}$$

$$SE = 16.3802 * \sqrt{0.225} = 16.3802 * 0.47434 = 7.7698 \text{ thousand dollars}$$

Step 4: Calculate the Margin of Error.

$$ME = t_{\alpha/2, 16} * SE = 2.120 * 7.7698 = 16.4720 \text{ thousand dollars}$$

Step 5: Compute the Confidence Limits.

$$\text{Point Estimate} = \bar{x}_1 - \bar{x}_2 = 320 - 305 = 15 \text{ thousand dollars}$$

$$\text{Lower Limit} = 15 - 16.4720 = -1.4720 \text{ thousand dollars } (-\$1,472)$$

$$\text{Upper Limit} = 15 + 16.4720 = 31.4720 \text{ thousand dollars } (\$31,472)$$

Conclusion: We are 95% confident that the true difference between the mean valuations of the two groups lies between -\$1,472 and \$31,472. Since this interval contains zero, there is no statistically significant difference between the two appraisal models.

PROBLEM 9: CONFIDENCE INTERVAL FOR THE DIFFERENCE BETWEEN TWO PROPORTIONS

A public health team evaluates the effectiveness of two distinct smoking-cessation initiatives across different cities.

Program 1: $n_1 = 200$ individuals tracked, $x_1 = 44$ successfully quit smoking.

Program 2: $n_2 = 250$ individuals tracked, $x_2 = 35$ successfully quit smoking.

Construct a 95% confidence interval for the true difference between the success rates of the two programs ($P_1 - P_2$).

Solution:

Step 1: Compute sample proportions.

$$p_1 = 44 / 200 = 0.22 \Rightarrow q_1 = 1 - 0.22 = 0.78$$

$$p_2 = 35 / 250 = 0.14 \Rightarrow q_2 = 1 - 0.14 = 0.86$$

$$\text{Point Estimate for the difference} = p_1 - p_2 = 0.22 - 0.14 = 0.08$$

Step 2: Determine the critical Z-value.

For a 95% confidence level, $Z_{\alpha/2} = 1.96$.

Step 3: Calculate the Standard Error of the Difference of Proportions.

$$SE = \sqrt{[(p_1 * q_1 / n_1) + (p_2 * q_2 / n_2)]}$$

$$SE = \sqrt{[(0.22 * 0.78 / 200) + (0.14 * 0.86 / 250)]}$$

$$SE = \sqrt{[(0.1716 / 200) + (0.1204 / 250)]}$$

$$SE = \sqrt{[0.000858 + 0.0004816]} = \sqrt{0.0013396} = 0.03660$$

Step 4: Calculate the Margin of Error.

$$ME = Z_{\alpha/2} * SE = 1.96 * 0.03660 = 0.07174$$

Step 5: Compute the Confidence Limits.

$$\text{Lower Limit} = 0.08 - 0.07174 = 0.00826 \text{ (or 0.83\%)}$$

$$\text{Upper Limit} = 0.08 + 0.07174 = 0.15174 \text{ (or 15.17\%)}$$

Conclusion: We are 95% confident that the true difference between the success rates of Program 1 and Program 2 is between 0.83% and 15.17%. Because the entire range is above zero, Program 1 is statistically more effective than Program 2.